

Coulomb's law

States that the force $F = k \frac{q_1 q_2}{r^2}$, where q_1 and q_2 are the magnitudes of the point charges that are separated by a distance r in a vacuum. k is Coulomb's constant.

The alternative way of expressing the equation is by replacing k with $\frac{1}{4\pi\epsilon_0}$;

$$F = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r^2} = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$$

Constant ϵ_0 is known as the **permittivity of free space** ('free space' means a vacuum).

The equation for Coulomb's law, above, only applies when the charges are in **a vacuum** (or in air, which has a permittivity $\approx$ that of a vacuum).

When charges are in a medium other than a vacuum or air, ϵ_0 must be replaced by $\epsilon_0 \epsilon_r$ where ϵ_r is the **relative permittivity** of the medium. Hence, Coulomb's law would be:

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 \epsilon_r r^2}$$

Relative permittivity is a measure of how easily the molecules that make up the substance can be rotated in an electric field. Relative permittivity has no units as it is a ratio.

Coulomb's law can predict the **direction of the electric force**; when both q_1 and q_2 are positive, the force F is also positive. The force must **repel** in such a situation.

Hence, if (at most) one of the two charges is negative, F is negative; the law predicts that the charges are **attracted**.